Multiplying by a Single Term

A monomial can have two parts: the coefficient and the variables. When you multiply two monomials, multiply each part separately. First multiply the coefficients. Then multiply each variable.

Use the rules of exponents to multiply variables. Remember, when you multiply variables with the same bases, keep the base the same and add the exponents.

Example 1

Hide

Multiply: $2x^3y \cdot 7xy^2$

Step 1
Multiply the coefficients.
$2x^3y \cdot 7xy^2$
$2 \cdot 7 = 14$

Step 2
Multiply the variables.
$x^3 \cdot x = x^{3+1} = x^4$
$y \cdot y^2 = y^{1+2} = y^3$

Step 3
Write the simplified expression.
$14x^4y^3$

You can also multiply an expression by a monomial. Use the distributive property to multiply each term in the expression by the monomial.

Example 2

Hide

Multiply: $7(z^3 + 5)$

Step 1
Use the distributive property.
$7(z^3 + 5)$
$(7 \cdot z^3) + (7 \cdot 5)$

Step 2
Multiply each pair of monomials.
$7 \cdot z^3 = 7z^3$
$7 \cdot 5 = 35$

Step 3
Write the expression.
$7z^3 + 35$

Example 3

Hide

Multiply: $-3ab(4a^2 + ab)$

Step 1
Use the distributive property.
$-3ab(4a^2 + ab)$
$(-3ab \cdot 4a^2) + (-3ab \cdot ab)$

Step 2
Multiply.
$-3ab \cdot 4a^2 = -12a^3b$
$-3ab \cdot ab = -3a^2b^2$

Step 3
Write the expression.
$-12a^3b - 3a^2b^2$

© 2025 New Readers Press.